Examiner only

SECTION A

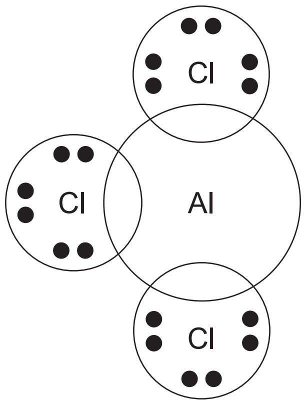
Answer **all** *questions in the spaces provided.*

1. Complete the electronic structure of the Co^{2+} ion below. [1]

$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$ ....

2. The compound $AlCl_3.NH_3$ contains both covalent and coordinate bonds.

Complete the dot and cross diagram for this compound. [2]



3. Write the expression for the ionic product of water, K_w. [1]

4. The standard hydrogen electrode and hydrogen fuel cells both use the same metal as an electrode. Name the metal and explain why this metal is suitable. [1]

....

....

Examiner only

5. When CCl_4 is added to water two separate layers are formed and there is no reaction, whilst $SiCl_4$ reacts violently with water.

Give the reason for this difference. [1]

....

....

....

6. The entropy change of the reaction shown below is positive. Give a reason for this. [1]

$$Na_2CO_3(aq) + 2HCl(aq) \longrightarrow 2NaCl(aq) + CO_2(g) + H_2O(l)$$

....

....

7. Carbon monoxide can be used to extract iron by the reduction of Fe_3O_4.

(a) Write an equation for this reaction. [1]

....

(b) Give a reason why carbon monoxide acts as a reducing agent while the equivalent oxide of lead does not. [1]

....

....

8. Draw the **structure** of the copper-containing ion formed when concentrated hydrochloric acid is added to a solution containing $[Cu(H_2O)_6]^{2+}$ ions. [1]

10

1410U301 03